

Executive Summary

Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor

Step 1. Project Objective

The objective of this project was to analyze the factory-to-customer shipping operations of Nassau Candy Distributor and transform raw logistics data into meaningful business insights. The project focused on evaluating shipping performance, product sales, factory efficiency, customer distribution, and overall business performance through data analysis and interactive dashboards

Step 2: Data Collection & Understanding

- Collected the Nassau Candy Distributor shipping dataset.
- Reviewed the dataset structure and business requirements.
- Loaded the dataset into Python using Pandas.
- Examined the dataset using `head()`, `info()`, `describe()`, and `shape()` to understand data types, missing values, and overall quality.

Output:

- Raw dataset ready for preprocessing.

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Step 3: Data Cleaning using Python

Imported the dataset into Python using Pandas and performed data cleaning.

Tasks performed:

- Removed duplicate records.
- Checked missing values.
- Treated missing data appropriately.
- Converted Order Date and Ship Date into datetime format.
- Removed invalid records.
- Standardized text columns (State, Region, Product Name).
- Checked incorrect data types.
- Verified Sales, Cost and Gross Profit values.

Output

- Clean and consistent dataset.

Step 4: Feature Engineering

Created new analytical columns required for logistics analysis.

New features created:

Shipping Lead Time

Shipping Lead Time = Ship Date – Order Date

Measured shipping duration in days.

Created additional fields like:

- Order Month
- Order Year
- Shipping Days
- Factory Name (mapped from Product Name)
- Route (Factory → Customer State)
- Route (Factory → Customer Region)

Output

Dataset enriched with logistics-related features.

Step 5: Exploratory Data Analysis (EDA)

Performed exploratory data analysis to understand shipment patterns.

Analysis included:

- Distribution of Shipping Lead Time
- Sales Analysis
- Profit Analysis
- Units Sold
- Ship Mode Distribution
- Region-wise Orders
- State-wise Orders
- Factory-wise Shipments
- Product Division Analysis
- Monthly Order Trend
- Correlation Analysis
- Outlier Detection

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Used visualizations such as:

- Bar Charts
- Line Charts
- Histograms
- Boxplots
- Heatmaps
- Pie Charts

Output

Identified shipping trends, regional demand and operational patterns

Step 6: Route Efficiency Analysis

Analyzed factory-to-customer shipping performance.

Calculated:

- Average Lead Time
- Route Volume
- Route Efficiency
- Fastest Routes
- Slowest Routes
- Delay Frequency

Compared performance across:

- Factory
- Region
- State
- Ship Mode

Output: Identified efficient and bottleneck shipping routes.

Step 7: Export Cleaned Dataset

After completing preprocessing and feature engineering:

- Exported the cleaned dataset as (cleaned_shipping_data.csv)

This dataset was used for visualization and dashboard development.

Output: Production-ready dataset.

Step 8: Tableau Dashboard Development

Imported the cleaned dataset into Tableau.

Created interactive dashboards including:

- Shipping Lead Time Analysis
- Factory-wise Performance
- State-wise Shipping Map
- Region Performance
- Ship Mode Comparison
- Monthly Shipment Trend
- KPI Cards
- Filters for State, Region, Ship Mode and Date

Output

Interactive business intelligence dashboard.

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Step 9: Streamlit Dashboard Development

Loaded the cleaned CSV into Streamlit.

Implemented interactive modules:

- KPI Cards
- Filters
- Route Efficiency Dashboard
- Lead Time Analysis
- Ship Mode Comparison
- Geographic Insights
- State Performance
- Charts and Tables

Users can dynamically explore shipping performance.

Output

Interactive analytics web application.

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Step 10: Business Insights

Generated actionable insights such as:

- Fastest factory-to-customer routes.
 - Regions experiencing maximum delays.
 - Best-performing shipping mode.
 - High-volume states with poor delivery performance.
 - Opportunities for logistics optimization.
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Step 11: Recommendations

Suggested improvements including:

- Optimize routes with consistently high lead times.
- Increase use of faster shipping modes on delay-prone routes.
- Improve warehouse allocation for high-volume regions.
- Monitor shipping KPIs regularly through dashboards.
- Use data-driven planning for logistics operations.

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Step 12: Conclusion

This project successfully transformed raw shipping data into meaningful business insights through data cleaning, feature engineering, exploratory data analysis, route efficiency evaluation, and interactive dashboards developed in Tableau and Streamlit. The findings support data-driven logistics decisions, helping improve shipping performance and operational efficiency.

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Overall Project Workflow

Raw Dataset

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Data Collection & Understanding

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Data Cleaning (Python)

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Missing Values Handling

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Duplicate Removal

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Datetime Conversion

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Feature Engineering (Shipping Lead Time, Factory Mapping, Routes)

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Exploratory Data Analysis (EDA)

